

2^K factorial designs

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Factor, levels and treatment combinations

- ▶ Factors: $F_1, F_2, \dots, F_K$.
- ▶ Each has two levels: 0 and 1.
- ▶ 2^K level combinations (treatments)

$$(0, 0, \dots, 0), \dots (1, 1, \dots, 1).$$

2^3 experiment

- ▶ Three factors: F_1 , F_2 , F_3 .
- ▶ Eight treatment combinations in lexicographic order shown below:

Table: Treatment combinations and potential outcomes for unit i in a 2^3 experiment

Treatment number (j)	Level combination	Potential outcome
1	000	$Y_i(1)$
2	001	$Y_i(2)$
3	010	$Y_i(3)$
4	011	$Y_i(4)$
5	100	$Y_i(5)$
6	101	$Y_i(6)$
7	110	$Y_i(7)$
8	111	$Y_i(8)$

Real-life experiments

AUDIT EXPERIMENT

Collaborators



Nicole Pashley (Rutgers U)



Brian Libgober (Northwestern U)

Racial discrimination to justice?

8/1/23, 12:35 PM

Why the US needs black lawyers even more than it needs black police | Race | The Guardian



Race

Why the US needs black lawyers even more than it needs black police

According to the American Bar Association, 88% of all lawyers are white and only 4.8% are black. This shouldn't matter in principle - until it does

Yolanda Young

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Mon 11 May 2015 14:45 EDT

In response to last year's killing of Michael Brown in Ferguson, La June Montgomery Tabron, who heads WK Kellogg Foundation, one of the US's largest philanthropies, issued a statement in which she astutely noted that deaths like Brown's "demonstrate that the law enforcement and justice systems in our nation are broken".

Yet as the tally of killings continues to climb, protests have mainly focused on police with an often-cited criticism: the force is too white.

Well, the legal profession is even whiter, and the job much more subjective.

Why audit experiments?

- ▶ Question: Do black and white individuals face equal barriers in acquiring legal services from lawyers?
- ▶ Surveys and interviews can induce social desirability bias.
- ▶ Audit experiments can be used to test for discriminatory behavior in such situations.
- ▶ Known to make strong causal claims; increasingly popular.

The audit application/instrument

Looking for a Lawyer

I am writing to ask if you would be willing to be my lawyer in a personal injury case. I was in a car accident a few weeks ago and had to go to the emergency room. Since the accident I haven't been able to go to work either. I'm hoping you can help me get the most I can out of insurance.

Sincerely,
{full name}

- ▶ Researchers assign “signals” (the treatment) to applications (sent to subjects) at random and compare success rates across treatment conditions. Here a name (e.g., Lakisha/Emily) highly correlated with the race is the signal.

Confounding of signals and the need for factorial experiment

- ▶ Names that are predictive of race are also correlated with other factors, e.g., gender, socio-economic status, age, religion.
- ▶ Solution: Use **factorial experiments** - apply multiple interventions or level combinations of factors on a population of experimental units or subjects (e.g., lawyers) using a randomized assignment mechanism.

Looking for a Lawyer

I am writing to ask if you would be willing to be my lawyer in a personal injury case. I was in a car accident a few weeks ago and had to go to the emergency room. Since the accident I haven't been able to go to work either (I make **{income}**). I'm hoping you can help me get the most I can out of insurance.

Sincerely,
{full name}

An experiment reported in Libgober (2020)

- ▶ Experimental units: 96 lawyers in private practice, selected from the California bar directory.
- ▶ Requests for representation from potential clients who claimed to be wrongfully accused of driving under the influence (DUI) - sent to lawyers by email.
- ▶ Three treatments/signals: Race - black (0) /white (1), Gender - female (0) /male (1), Income - low (0) /high (1).
- ▶ Eight treatment combinations 1:000, 2:001, 3:010, 4:011, 5:100, 6:101, 7:110, 8:111.
- ▶ A balanced 2^3 factorial experiment with 12 units per treatment combination.
- ▶ Response: Binary, taking value 1 if the lawyer replied in any fashion at all and 0 otherwise.

Potential outcomes and estimands in audit experiment

Unit (i)	Level Combinations				Factorial effect (Race)
	1: 000	2: 001	...	8: 111	
1	$Y_1(1)$	$Y_1(2)$	...	$Y_1(8)$	$\tau_{1,R}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
96	$Y_{96}(1)$	$Y_{96}(2)$	...	$Y_{96}(8)$	$\tau_{96,R}$
Average	$\bar{Y}(1)$ $= P_1$	$\bar{Y}(2)$ $= P_2$	...	$\bar{Y}(8)$ $= P_8$	τ_R
Variance	S_1^2	S_2^2	...	S_8^2	$S_{\tau_R}^2$

- ▶ *Only one* out of 8 potential outcomes will be observed for each unit.
- ▶ Estimand:

$$\tau_R = (P_5 + P_6 + P_7 + P_8)/4 - (P_1 + P_2 + P_3 + P_4)/4$$
 (and six other factorial effects).

Summary of observed outcomes

Unit (i)	Potential Outcomes							
	1: 000	2: 001	3: 010	4: 011	5: 100	6: 101	7: 110	8: 111
1	1	?	?	?	?	?	?	?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
96	?	?	0	?	?	?	?	?
Prop P	P_1	P_2	P_3	P_4	P_5	P_6	P_7	P_8
Measure	Observed Outcomes							
	1: 000	2: 001	3: 010	4: 011	5: 100	6: 101	7: 110	8: 111
N_j	12	12	12	12	12	12	12	12
resp	2	2	2	3	5	2	5	6
non-resp	10	10	10	9	7	10	7	6
$\hat{P}$	0.167	0.167	0.167	0.250	0.417	0.167	0.417	0.500
variance	0.152	0.152	0.152	0.205	0.265	0.152	0.265	0.273

- Conversion of raw data to this form?

Estimating the main effect of race

Unit (i)	Potential Outcomes							
	1: 000	2: 001	3: 010	4: 011	5: 100	6: 101	7: 110	8: 111
1	1	?	?	?	?	?	?	?
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
96	?	?	0	?	?	?	?	?
Prop P	P_1	P_2	P_3	P_4	P_5	P_6	P_7	P_8
Measure	Observed Outcomes							
	1: 000	2: 001	3: 010	4: 011	5: 100	6: 101	7: 110	8: 111
N_j	12	12	12	12	12	12	12	12
resp	2	2	2	3	5	2	5	6
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$\hat{P}$	0.167	0.167	0.167	0.250	0.417	0.167	0.417	0.500
variance	0.152	0.152	0.152	0.205	0.265	0.152	0.265	0.273

► Estimand: $\tau_R = (P_5 + P_6 + P_7 + P_8)/4 - (P_1 + P_2 + P_3 + P_4)/4$.

► Estimator:

$$\hat{\tau}_R = (\hat{P}_5 + \hat{P}_6 + \hat{P}_7 + \hat{P}_8)/4 - (\hat{P}_1 + \hat{P}_2 + \hat{P}_3 + \hat{P}_4)/4 = 0.19$$

(verify!)

Exercise: Estimate the main effect of gender

Unit (i)	Potential Outcomes							
	1: 000	2: 001	3: 010	4: 011	5: 100	6: 101	7: 110	8: 111
1	1	?	?	?	?	?	?	?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
96	?	?	0	?	?	?	?	?
Prop P	P_1	P_2	P_3	P_4	P_5	P_6	P_7	P_8
Measure	Observed Outcomes							
	1: 000	2: 001	3: 010	4: 011	5: 100	6: 101	7: 110	8: 111
N_j	12	12	12	12	12	12	12	12
resp	2	2	2	3	5	2	5	6
non-resp	10	10	10	9	7	10	7	6
$\hat{P}$	0.167	0.167	0.167	0.250	0.417	0.167	0.417	0.500
variance	0.152	0.152	0.152	0.205	0.265	0.152	0.265	0.273

► Estimand: $\tau_G = ?$.

► Estimator: $\hat{\tau}_G = ?$.

Exercise: Estimate the interaction effect between race and gender

Unit (i)	Potential Outcomes							
	1: 000	2: 001	3: 010	4: 011	5: 100	6: 101	7: 110	8: 111
1	1	?	?	?	?	?	?	?
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
96	?	?	0	?	?	?	?	?
Prop P	P_1	P_2	P_3	P_4	P_5	P_6	P_7	P_8
Measure	Observed Outcomes							
	1: 000	2: 001	3: 010	4: 011	5: 100	6: 101	7: 110	8: 111
N_j	12	12	12	12	12	12	12	12
resp	2	2	2	3	5	2	5	6
non-resp	10	10	10	9	7	10	7	6
$\hat{P}$	0.167	0.167	0.167	0.250	0.417	0.167	0.417	0.500
variance	0.152	0.152	0.152	0.205	0.265	0.152	0.265	0.273

► Estimand: $\tau_{R \times G} = ?$.

► Estimator: $\hat{\tau}_{R \times G} = ?$.

Let's visualize these effects

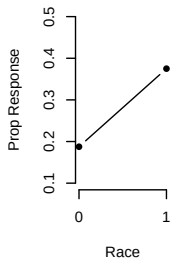
Measure	Observed Outcomes							
	1: 000	2: 001	3: 010	4: 011	5: 100	6: 101	7: 110	8: 111
N_j	12	12	12	12	12	12	12	12
resp	2	2	2	3	5	2	5	6
non-resp	10	10	10	9	7	10	7	6
$\hat{P}$	0.167	0.167	0.167	0.250	0.417	0.167	0.417	0.500
variance	0.152	0.152	0.152	0.205	0.265	0.152	0.265	0.273

Race	gender		Avg
	0	1	
0	0.167	0.201	0.184
1	0.292	0.459	0.376
Avg	0.223	0.329	0.28

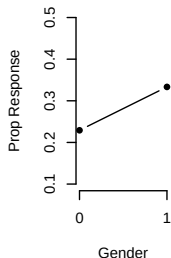
Let's visualize these effects (contd).

Race	gender		Avg
	0	1	
0	0.167	0.201	0.184
1	0.292	0.459	0.376
Avg	0.223	0.329	0.28

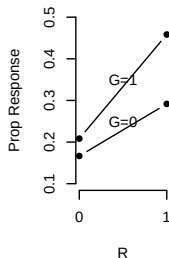
M.E.(Race)



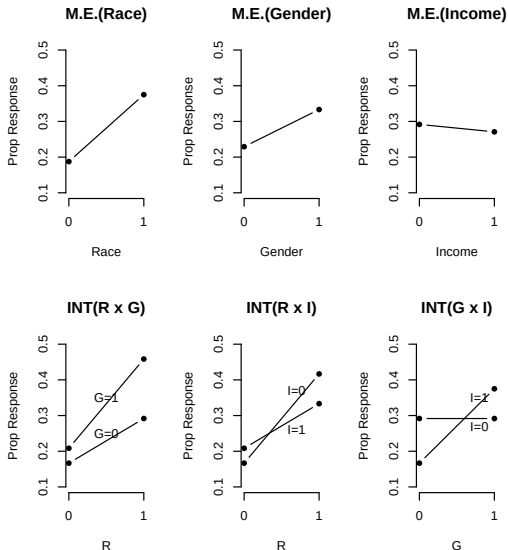
M.E.(Gender)



INT(R x G)



All MEs and 2FIs



Calculation of all factorial effect estimates

```
p <- c(0.1667,0.1667,0.1667,0.2500,0.4167,0.1667,0.4167,0.5000)
s_sq <- c(0.1515,0.1515,0.1515,0.2045,0.2652,0.1515,0.2652,0.2727)
```

```
D = gen.factorial(2,3) ## ME contrast columns in reverse order
D[, c(1:3)] = D[,c(3:1)] ## Reverse the order of the columns
colnames(D) = c("R", "G", "I")
L = model.matrix(~ R + G + I + R*G + R*I + G*I + R*G*I,D)
```

```
tau_hat <- (t(L) %*% p)/4 # Factorial effects
```

```
> tau_hat
```

	[,1]
(Intercept)	0.56255
R	0.18750
G	0.10415
I	-0.02085
R:G	0.06250
R:I	-0.06250
G:I	0.10415
R:G:I	0.06250

Fisher's sharp null hypothesis

For a 2^3 experiment, Fisher's sharp null hypothesis of no treatment effect on any unit can be written as:

$$H_0 : Y_i(1) = Y_i(2) = \dots = Y_i(8), \quad i = 1, \dots, N,$$

or equivalently as

$$\tau_i(F_1) = \tau_i(F_2) = \dots = \tau_i(F_1 F_2 F_3) = 0, \quad i = 1, \dots, N.$$

that is, all the J unit-level factorial effects are equal for each unit.

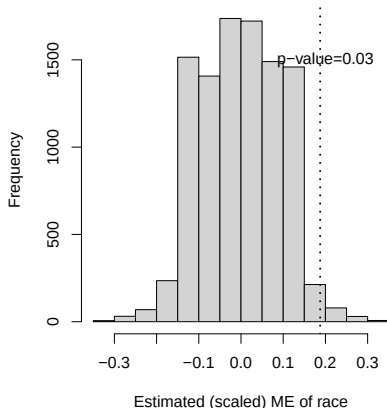
Test statistic

- ▶ Each of the seven estimated factorial effects can be used as a test statistic because each defines a specific type of deviation from the null hypothesis.
- ▶ Therefore, a Fisher randomization test can be obtained by comparing each estimated factorial effect against its randomization distribution under the sharp null hypothesis of no treatment effects.
- ▶ Due to the symmetry of the definition of all factorial effects, the reference distribution under the sharp null hypothesis of no effects is the same for all seven factorial effects.

Fisherian inference of factorial effects

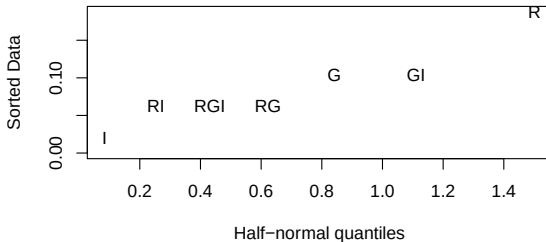
- ▶ Sharp null hypothesis of no treatment effect on any unit.
- ▶ Randomization test using scaled estimated treatment effect of race (dividing each estimated factorial effect by it's estimated standard

error $\sqrt{\frac{1}{2^{2(K-1)}} \sum_{j=1}^8 \frac{s_j^2}{N_j}} = 0.092$.



Visualization to assess significance: half normal plots

```
-----  
effects = c("R","G","I", "RG", "RI", "GI", "RGI")  
estimates = as.numeric(tau_hat[2:8])  
factorial = data.frame(effects, estimates)  
  
library(faraway)  
halfnorm(factorial$estimates, nlab=7, labs=factorial$effects)  
-----
```



Half normal plots: the idea

- ▶ Under Fisher's sharp null hypothesis of no effect on any unit, each factorial effect $\bar{\tau}(\cdot)$ is zero and each column of the potential outcomes matrix has same variance.
- ▶ By central limit theorem (finite population), each estimated factorial effect $\hat{\tau}(\cdot)$ is approximately normal with the same mean and variance. Further, if the design is balanced, covariance between any pair of estimated effects is zero.
- ▶ Thus, for a balanced design (which is the case here) the estimated (absolute) factorial effects can be considered as iid observations from a *half-normal* distribution from the same mean (zero) and variance under the sharp null hypothesis of no treatment effect on any unit.
- ▶ Half normal plots can be applied sequentially, by removing the significant effects at each stage.

Neymanian inference

- ▶ Pashley, N., Libgober, B. and Dasgupta, T.
“Analysis and sample-size determination for 2^K audit experiments with binary response and application to identification of effect of racial discrimination on access to justice”, to appear in the *Journal of the Royal Statistical Society, Series A*.
- ▶ Previous version available here:
<https://arxiv.org/abs/2503.10591>

Food for thought

- ▶ Suppose 48 of the 96 subjects (lawyers) were female and the remaining were male. Would you use the same design (i.e., completely randomized allocation of the 96 lawyers into the eight treatment groups)? If not, how would you change your design? How would your analysis (Fisherian inference) change?

The New York School Experiment

- ▶ Dasgupta, Pillai and Rubin (2015), Branson, Dasgupta and Rubin (2016).
- ▶ 224 New York schools.
- ▶ Five new interventions labelled A-E, e.g., Quality review (A), School-wide performance bonus scheme for the teachers (B).
- ▶ Each school can receive an active (treatment) or a passive (control) level of each intervention. Thus, there are 2^5 possible treatment combinations.
- ▶ Response: A cumulative score on the annual progress report.
- ▶ Each school has at least 50 pre-treatment covariates (e.g. ethnicity distribution of students, attendance rate of teachers).